

SANTIAGO DE CUBA, CubaWMO: **782640**Lat: **19.970N**Lon: **75.835W**Elev: **76**StdP: **100.42**Time Zone: **-5.00 (CUB)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	19.8	12.0	8.8	22.9	13.3	9.6	22.9	9.5	24.0	8.6	24.3	4.2	0	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	32.1	24.9	31.8	25.0	31.2	25.0	27.2	29.7	26.7	29.6	26.3	29.5	4.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.5	28.9	26.0	21.6	28.7	25.4	20.7	28.5	86.6	29.6	84.2	29.7	82.6	29.6	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.0	6.2	17.2	34.8	1.2	1.7	16.3	36.1	15.6	37.1	14.9	38.0	14.0	39.3		
(5)			14.2	27.7	1.3	1.0	13.2	28.4	12.5	29.0	11.7	29.6	10.7	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	24.4	24.6	24.9	25.6	26.3	27.1	27.7	27.9	27.6	27.0	26.2	25.3		
	DBStd	1.53	1.21	1.10	1.07	0.84	0.89	0.80	0.68	0.75	0.79	0.89	1.09	1.20		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	5922	448	409	461	468	506	513	548	556	528	527	485	474		
	CDD18.3	2881	189	176	203	218	248	263	289	297	278	269	235	216		
	CDH23.3	26218	1235	1255	1514	1797	2290	2748	3196	3326	2944	2521	1859	1533		
	CDH26.7	8144	242	241	298	413	650	926	1205	1286	1093	835	548	407		
(14) Wind	WSAvg	3.1	3.6	3.5	3.6	3.3	2.9	2.7	2.8	2.9	2.8	2.8	3.3	3.4		
(15) Precipitation	PrecAvg	1124	41	40	55	76	179	93	81	91	121	187	96	40		
	PrecMax	2557	248	196	151	267	482	260	253	229	483	1360	320	133		
	PrecMin	638	1	0	3	0	6	10	5	37	24	38	7	0		
	PrecStd	382	56	44	41	54	129	59	49	42	81	224	70	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	30.2	30.3	30.2	31.2	31.8	32.2	33.1	32.8	32.1	31.8	31.1		
		MCWB	21.8	21.7	21.9	23.3	24.6	25.2	25.0	24.9	24.7	24.7	23.6	23.0		
	2%	DB	29.2	29.2	29.2	29.8	30.2	31.1	31.9	32.1	31.9	31.2	30.9	30.1		
		MCWB	22.2	22.2	22.6	23.4	24.5	25.2	25.1	25.2	25.2	25.0	23.9	23.1		
	5%	DB	28.8	28.8	28.9	29.1	30.0	30.8	31.2	31.6	31.2	30.9	30.1	29.4		
		MCWB	22.2	22.3	22.6	23.3	24.4	25.2	25.2	25.2	25.2	25.0	24.0	22.9		
	10%	DB	28.1	28.1	28.2	28.8	29.2	30.1	30.9	31.1	30.9	30.2	29.5	28.9		
		MCWB	22.2	22.2	22.3	23.2	24.2	25.1	25.2	25.3	25.2	24.9	23.9	22.9		
	0.4%	WB	25.5	26.0	26.3	26.5	27.2	27.7	27.4	27.4	27.6	27.3	27.0	26.2		
		MCDB	27.8	27.7	28.1	28.4	29.0	29.4	30.4	30.4	29.8	29.4	28.9	28.8		
	2%	WB	24.4	24.6	25.0	25.4	26.2	26.9	26.7	26.7	26.7	26.5	26.1	25.2		
		MCDB	27.3	27.2	27.6	28.1	28.6	29.2	30.1	30.3	30.0	29.4	28.7	28.2		
	5%	WB	23.6	23.7	24.0	24.6	25.7	26.3	26.2	26.2	26.2	26.0	25.5	24.4		
		MCDB	27.0	26.9	27.3	27.9	28.4	29.1	29.9	30.0	29.7	29.1	28.4	27.7		
	10%	WB	22.9	23.1	23.4	24.0	25.2	25.9	25.9	26.0	25.9	25.6	24.8	23.7		
		MCDB	26.9	26.8	27.0	27.6	28.2	29.0	29.7	29.8	29.5	28.9	28.1	27.2		
(35) Mean Daily Temperature Range	MDBR		7.7	7.9	7.5	6.8	6.5	6.5	7.0	7.0	7.1	6.9	7.3	7.8		
	5% DB	MCDBR	8.4	8.3	7.9	7.2	6.9	7.1	7.3	7.6	7.5	7.6	7.9	8.4		
		MCWBR	4.2	4.2	3.9	3.7	3.6	3.8	3.9	3.7	3.6	3.4	3.6	4.0		
	5% WB	MCDBR	7.5	7.3	6.5	6.4	6.0	6.3	6.8	6.8	6.8	6.5	6.7	7.4		
		MCWBR	4.6	4.5	4.0	4.0	3.9	4.0	4.1	3.8	3.8	3.7	3.9	4.3		
(40) Clear-Sky Solar Irradiance	taub		0.374	0.380	0.386	0.422	0.449	0.484	0.492	0.462	0.424	0.401	0.388	0.382		
	taud		2.520	2.494	2.479	2.388	2.343	2.232	2.194	2.312	2.437	2.508	2.505	2.513		
	Ebn at Noon		903	917	921	886	852	815	809	839	873	885	882	882		
	Edh at Noon		99	106	112	124	128	142	148	132	115	104	99	97		
(44) All-Sky Solar Radiation	RadAvg		4.50	5.35	5.98	6.32	6.04	6.06	6.35	6.15	5.70	4.88	4.50	4.26		
	RadStd		0.29	0.29	0.29	0.33	0.36	0.37	0.31	0.20	0.37	0.41	0.32	0.26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	+1.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page